

The Zero-Beta Interest Rate (2023)

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BGGMP Reading Group

Motivation

- Interest rates are important.
 - ▶ Intertemporal price of goods
 - ▶ Central bank manage business cycles

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- ▶ Traditionally, use risk-free nominal bond: $R_{t+1} = R_{b,t}$
- ▶ Problem:
 - ★ Euler equation does not hold for Treasury bill return
 - ★ There are many safe assets: cash, deposits, treasuries, commercial paper
 - ★ Safe assets have non-pecuniar convenience

Motivation

- This paper:
 - ▶ Consumption Euler equation should hold for asset with no convenience
 - ▶ In particular it holds for the zero beta rate:
 - ★ The expected return of a stock portfolio orthogonal to the SDF
 - ★ Zero beta rate is high and volatile
 - ▶ Applications:
 - ★ Monetary tightening lowers the zero beta rate
 - ★ (Risk premia are low)

Implications of aggregate consumption EE

- SDF: $\Lambda_t = \delta^t c_t^{-\sigma} P_t^{-1}$
- EE implies:

$$0 = \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} (R_{i,t+1} - R_{j,t+1}) \right] \quad (1)$$

$$R_{b,t}^{-1} = \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} \right] \quad (2)$$

$$R_{0,t}^{-1} = \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} \right] \quad (3)$$

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$$R_{0,t}^{-1} = \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} \right] \quad (3)$$

- (1) violated bc other factors necessary
- (2) violated bc it doesn't hold in time series and multiple safe assets
- Next: model where (1) and (2) fail but (3) holds

Model setup

- Representative household:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t \xi_t \left(\frac{c_t^{1-\sigma}}{1-\sigma} + \eta_{m,t} \log(M_t/P_t) + \eta_{b,t} \log(B_t/P_t) \right) \right]$$

- Budget constraint:

$$P_t c_t + (B_t - B_t^s) + M_t + \sum_{i=1}^N X_{i,t} \leq$$

$$(B_{t-1} - B_{t-1}^s) R_{b,t-1} + M_{t-1} + \sum_{i=1}^N X_{i,t-1} R_{i,t} + P_t y_t + M_t^s - M_{t-1}^s$$

- Market clearing: $c_t = y_t$, $B_t = B_t^s$, $X_{i,t} = 0$, $M_t = M_t^s$
- Do not model supply side

Rederive Euler equations

- SDF:

$$\Lambda_t = \delta^t \xi_t c_t^{-\sigma} / P_t$$

- Euler equations for money, safe bonds, and risky assets:

$$c_t^{-\sigma} = \eta_{m,t} (M_t / P_t)^{-1} + \delta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \frac{1}{P_{t+1} / P_t} \right]$$

$$c_t^{-\sigma} = \eta_{b,t} (B_t / P_t)^{-1} + \delta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \frac{R_{b,t}}{P_{t+1} / P_t} \right]$$

$$c_t^{-\sigma} = \delta \mathbb{E}_t \left[\frac{\xi_{t+1}}{\xi_t} c_{t+1}^{-\sigma} \frac{R_{i,t+1}}{P_{t+1} / P_t} \right]$$

- In traditional EE: $\eta_{m,t} = \eta_{b,t} = 0$ and $\xi_t = 1$

Rederive Euler equations

- Let $R_{p,t+1}$ return on portfolio uncorrelated with Λ_{t+1}/Λ_t , then

$$c_t^{-\sigma} = \delta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \frac{R_{p,t+1}}{P_{t+1}/P_t} \right] = \delta R_{0,t} \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{P_{t+1}/P_t} \right]$$

- ▶ where $R_{0,t} \equiv \mathbb{E}_t[R_{p,t+1}]$ is the **zero beta rate**

Measuring $R_{0,t}$: An infeasible but intuitive procedure

- Assume we know excess returns $R_{t+1} - R_{0,t}$
- 3 step procedure: Asset pricing assumptions
 - ① For each asset i , excess returns can be projected on factors to get α and β

$$R_{i,t+1} - R_{0,t} = \alpha_i + \sum_{j=1}^K \beta_{i,j} F_{j,t+1} + \varepsilon_{i,t+1}$$

- ② Using betas, choose a zero beta portfolio w : $w' \beta = 0$. Implies:
 $\mathbb{E}_t[w' (R_{t+1} - R_{0,t})] = 0$
- ③ Predict return with OLS on instruments:

$$\underbrace{w' R_{t+1}}_{R_{p,t+1}} = \underbrace{\gamma' Z_t}_{R_{0,t}} + u_{t+1}$$

- Sidenote: w is arbitrary as long as portfolio is zero beta, so choose w to minimize $\text{Var}(u_{t+1})$

Measuring $R_{0,t}$: GMM procedure

- Want to estimate $\theta = (\alpha, \beta, \gamma)$
- Modified procedure:
 - ▶ Moment condition step 1: $\mathbb{E}[\hat{\epsilon}_{t+1}(\theta) \otimes F_{t+1}(\theta)] = 0$
 - ▶ Zero beta portfolios have expected return equal to zero beta rate:
Define: $H(\theta) = I_N - \beta(\theta)\beta(\theta)^+$, then zero beta weight vector is $H(\theta)w$ for any w
 - ▶ Moment condition step 3: $\mathbb{E}[H(\theta) \cdot (R_{t+1} - \iota\gamma'(\theta) \cdot Z_t) \otimes Z_t] = 0$
- GMM: $\mathbb{E}[g_{t+1}(\theta)] = 0$

$$g_{t+1}(\theta) = \begin{bmatrix} \hat{\epsilon}_{t+1}(\theta) \otimes F_{t+1}(\theta) \\ H(\theta) \cdot (R_{t+1} - \iota\gamma'(\theta) \cdot Z_t) \otimes Z_t \end{bmatrix}$$

Measuring $R_{0,t}$: GMM procedure

- Given covariance matrix of excess returns $\Sigma_R(\theta)$, choose minimum variance zero beta portfolio:

$$\begin{aligned} \min_w & w' \Sigma_R(\theta) w \\ \text{s.t. } & w' \beta = \mathbf{0} \text{ and } w' \iota = 1 \end{aligned}$$

- Feasible and explicit solution for $w(\theta)$ Solution

- Then weight matrix:

$$W(\theta) = \begin{bmatrix} I_{N \times (K+1)} & \mathbf{0} \\ \mathbf{0} & w(\theta)w(\theta)' \otimes I_{L+1} \end{bmatrix}$$

Data

- Monthly January 1973 - December 2020
- Equity portfolio returns R_{t+1}
- Factors F_t
- Instruments/Predictors Z_t
- Consumption C_t

Data

- Monthly January 1973 - December 2020
- Equity portfolio returns R_{t+1}
 - ▶ CRSP equity returns matched to COMPUSTAT firms (excluding bottom 20% stocks by market value)
 - ★ Compute 5 year trailing beta to CRSP market return
 - ▶ 27 portfolios based on (i) beta, (ii) market value, (iii) book-to-market
 - ▶ 27 portfolios based on (i) beta, (ii) market value, (iii) investment rate
 - ▶ 27 portfolios based on (i) beta, (ii) market value, (iii) profitability
 - ▶ 49 industry (4-digit SIC) portfolios from Ken French
- Factors F_t
- Instruments/Predictors Z_t
- Consumption C_t

Data

- Monthly January 1973 - December 2020
- Equity portfolio returns R_{t+1}
- Factors F_t
 - ▶ 5 equity factors from Fama and French (2015)
 - ★ Excess market return ($R_{m,t} - R_{0,t}$)!
 - ★ SMB Small minus big market value
 - ★ HMA High minus low book to market stocks
 - ★ RMW Robust minus weak profitability
 - ★ CMA Low minus high investment rate
 - ▶ Return of 6-10y Treasury bond portfolio over a 1m Treasury bill
 - ▶ Return of long term corporate bond over long term treasury bond
- Instruments/Predictors Z_t
- Consumption C_t

Data

- Monthly January 1973 - December 2020
- Equity portfolio returns R_{t+1}
- Factors F_t
- Instruments/Predictors Z_t
 - ▶ 1m Treasury bill yield
 - ▶ 12 month rolling average of inflation
 - ▶ Term spread (10y less 3m Treasury yield)
 - ▶ Excess bond premium (EBP)
 - ▶ Unemployment rate
- Consumption C_t

Data

- Monthly January 1973 - December 2020
- Equity portfolio returns R_{t+1}
- Factors F_t
- Instruments/Predictors Z_t
- Consumption C_t
 - ▶ NIPA non-durable goods and services consumption per capita growth

Instruments predict zero beta portfolio returns

- Recall, want γ from: $w' \mathbf{R}_{t+1} = \gamma' Z_t + u_{t+1}$

Table 1: Constructing the Zero-Beta Rate

	(1)	(2)
	GMM	OLS (inf.)
RF	2.747 (0.706)	2.747 (0.686)
UMP	0.0629 (0.0820)	0.0629 (0.0818)
EBP	-0.993 (0.256)	-0.993 (0.238)
TSP	0.297 (0.0966)	0.297 (0.0958)
CPI_Rolling	-2.196 (0.909)	-2.196 (0.875)
Constant	1.002 (0.112)	1.002 (0.111)
Wald/F	35.74	8.125
p-value	1.1e-06	2.0e-07
RMSE		2.671
Observations	574	574

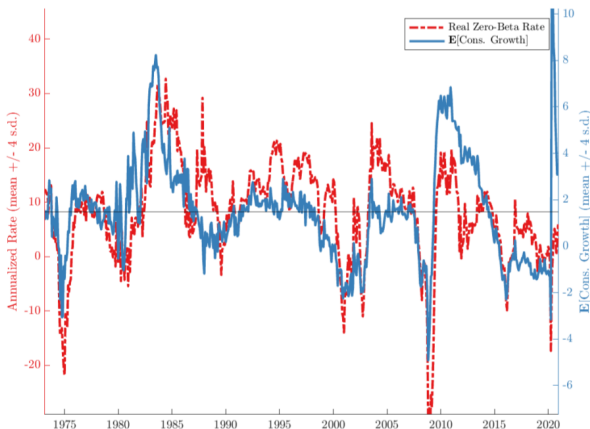
Standard errors in parentheses

- Zero beta rate falls when a recession is likely

Zero beta rate fits the Euler equation

- Linear EE:

- $\mathbb{E}_t[\log(c_{t+1}/c_t)] = \sigma^{-1} \log(\delta) + \sigma^{-1}(\log(R_{0,t}) - \mathbb{E}_t[\log(P_{t+1}/P_t)])$



- $\mathbb{E}_t[\log(c_{t+1}/c_t)]$ and $\mathbb{E}_t[\log(P_{t+1}/P_t)]$ from predicted reg on Z_t

Zero beta rate is high and volatile

- Average real zero beta rate 8.3% per year
 - ▶ Large and volatile spread over Treasury bill yield: 7.9% per year
 - ▶ Similar to average real market return 8.1% per year
- Standard deviation of $R_{p,t+1} - R_{0,t}$ is 9.4% annualized
 - ▶ Less volatile than market portfolio

Statistical test for Euler equation

- Test instrumented non-linear Euler equation:

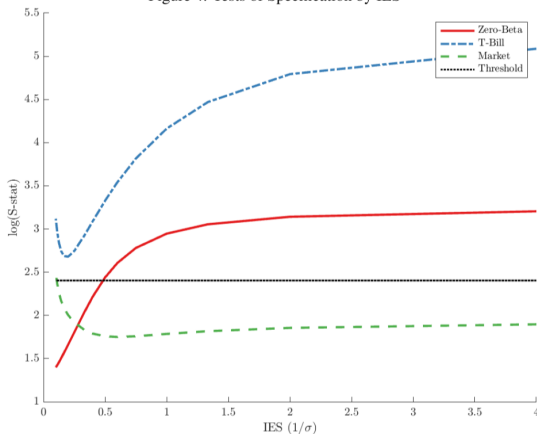
$$\mathbb{E} \left[\left(\delta \left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{R_{0,t}}{P_{t+1}/P_t} - 1 \right) Z_{l,t} \right] = 0$$

- Want to test whether for some σ^{-1} model cannot be rejected:
 - ① Conjecture value for σ
 - ② Estimate $\hat{\delta}(\sigma)$ using $Z_{0,t} = 1$ only
 - ③ Test using all instruments
- In GMM, $g_{Test,t}(\theta, \delta, \sigma)$ is vector of EE moments:

$$g_{l,t+1}(\theta, \delta, \sigma) = \left(\delta \frac{c_{t+1}^{-\sigma}}{c_t^{-\sigma}} \frac{R_{0,t}(\theta)}{P_{t+1}/P_t} - 1 \right) Z_{l,t}$$

EE cannot be rejected for IES below 0.5

Figure 4: Tests of Specification by IES



- Using Treasury bill, always reject EE
- Using market return, never reject EE (weak identification)

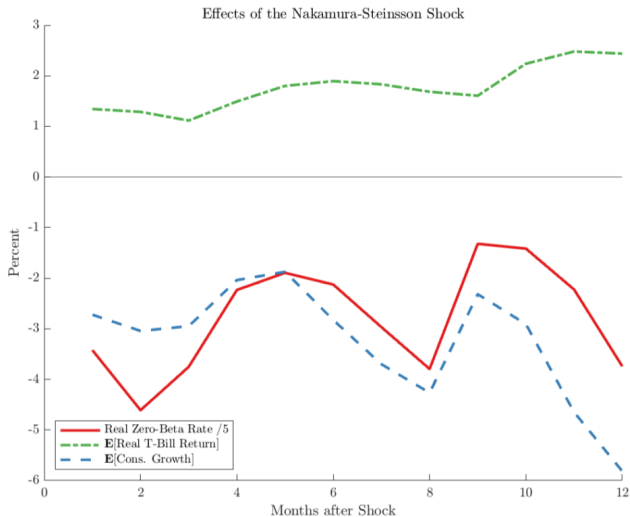
Monetary policy shocks

- Monetary policy exploits Euler equation
 - ▶ But large and time-varying spread between $R_{b,t}$ and $R_{0,t}$
- Run a regression:

$$\gamma_j(Z_{t+h} - Z_{t-1}) = \phi_{0,h} + \phi_{1,h}mpshock_t + \epsilon_{t+h}$$

- ▶ γ_j 's are the coefficients on the predictive regressions on consumption growth, real Treasury bill returns, and the real zero beta rate
- ▶ *mpshock* is either Romer and Romer (2004) or Nakamura and Steinsson (2018) shock

Monetary tightening lowers zero beta rate



Intuition

- Monetary tightening lowers expected consumption growth
 - ▶ interest rate falls as households correctly start saving
 - ▶ point estimate consistent with $\sigma = 5$ ($\text{IES} = 0.2$)
- Importantly, EE using Treasury bill return fails conditionally on monetary shock
 - ▶ Need habits (like Christiano et al 2005), additional wedges (like Smets and Wouters 2007), informational frictions (like Auclert et al 2020)
- If you accept the zero beta rate as correct intertemporal price of consumption, no need for these frictions

More slide

Conclusion

- Interest rate is central in macro and asset pricing
 - ▶ Conventional view: safe interest rate as intertemporal price
 - ★ Violates EE and generates large risk premia
 - ▶ This paper: zero beta rate is the correct intertemporal price
 - ★ EE holds (unconditionally and conditionally on monetary shocks) and risk premia are low
- Zero beta rate solves EE failure and risk premium puzzle but introduces "convenience puzzle"

Measuring $R_{0,t}$: Asset pricing assumptions Back

- Excess returns of assets can be projected on factors:

$$R_{i,t+1} - R_{0,t} = \alpha_i + \sum_{j=1}^K \beta_{i,j} F_{j,t+1} + \epsilon_{i,t+1}$$

- Assume betas are constant over time (use portfolios)
- Nonlinear SDF Λ_{t+1}/Λ_t can be approximated by linear factor structure
 - First factor: $F_{1,t+1} = R_{m,t+1} - R_{0,t}$ and other factors do not include zero beta rate or non-traded factors

$$\frac{\Lambda_{t+1}}{\Lambda_t} = R_{0,t}^{-1} + \sum_{j=1}^K \omega_{j,t} (F_{j,t+1} - \mathbb{E}_t[F_{j,t+1}]) + \zeta_{t+1}$$

- Thus nonlinear SDF is equivalent to linear SDF for pricing equities:

$$1 = \mathbb{E}_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} R_{i,t+1} \right] = \mathbb{E}_t \left[\left(R_{0,t}^{-1} + \sum_{j=1}^K \omega_{j,t} (F_{j,t+1} - \mathbb{E}_t[F_{j,t+1}]) \right) R_{i,t+1} \right]$$

Measuring $R_{0,t}$: Asset pricing assumptions [Back](#)

- Zero beta rate is linear in predictor variables $Z_{l,t}$. Assume

$$R_{0,t}(\gamma) = \gamma' Z_t$$

- Z_t includes Treasury bill yield $R_{b,t}$ so nest traditional view as special case

Minimum variance zero beta portfolio [Back](#)

- Feasible requires Σ_R full rank and ι does not lie in span of β

$$w(\theta) = \Sigma_R(\theta)^{-1} [\iota \quad \beta(\theta)] \left(\begin{bmatrix} \iota' \\ \beta(\theta)' \end{bmatrix} \Sigma_R(\theta)^{-1} [\iota \quad \beta(\theta)] \right)^+ \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}$$

- Use Ledoit and Wolf (2017) estimator for Σ_R

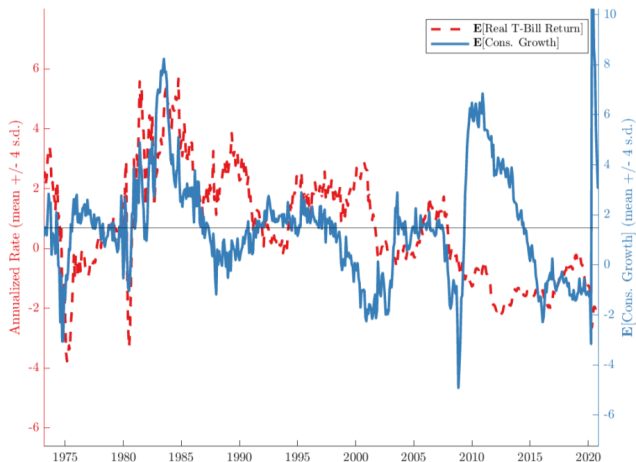
More data for robustness [Back](#)

- Additional factors
 - ▶ Consumption growth (but after controlling for other factors, consumption does not matter)
- Additional instruments
 - ▶ CAPE
 - ▶ Shadow spread between bill yields and implied yields by smooth term structure
 - ▶ Moody's BAA minus AAA corporate bond spread
 - ▶ Lagged consumption growth
 - ▶ Lagged inflation

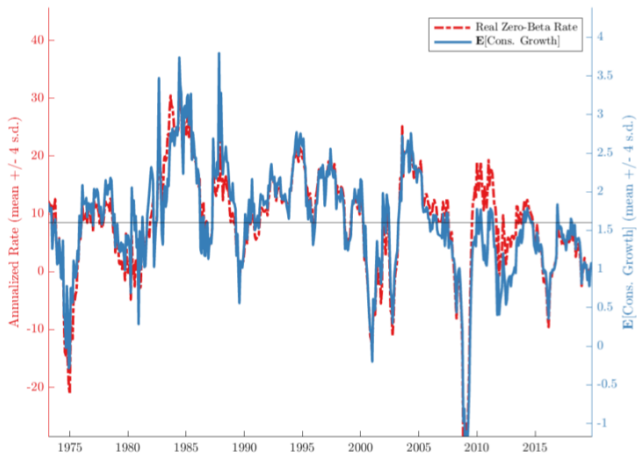
Table 2: Summary Statistics of Instruments and Consumption Growth

	Mean	SD	Min	25%	50%	75%	Max
T-bill Yield (% , Monthly)	0.370	0.288	0.000	0.110	0.390	0.538	1.350
Unemployment (%)	6.287	1.709	3.500	5.000	5.900	7.400	14.700
Excess Bond Premium (% , Annual)	0.075	0.549	-1.085	-0.263	-0.048	0.252	3.472
CAPE	20.761	8.861	6.639	13.015	20.844	26.495	44.198
Term Spread (% , Annual)	1.760	1.366	-3.040	0.750	1.920	2.797	6.940
Shadow Spread (% , Monthly)	0.023	0.021	-0.164	0.012	0.020	0.030	0.137
Corporate Bond Spread (% , Annual)	1.088	0.449	0.550	0.780	0.950	1.270	3.380
Rolling Inflation (% , Monthly)	0.315	0.213	0.050	0.174	0.221	0.380	1.069
Consumption Growth (% , Monthly)	0.124	0.747	-12.909	-0.037	0.139	0.315	6.150

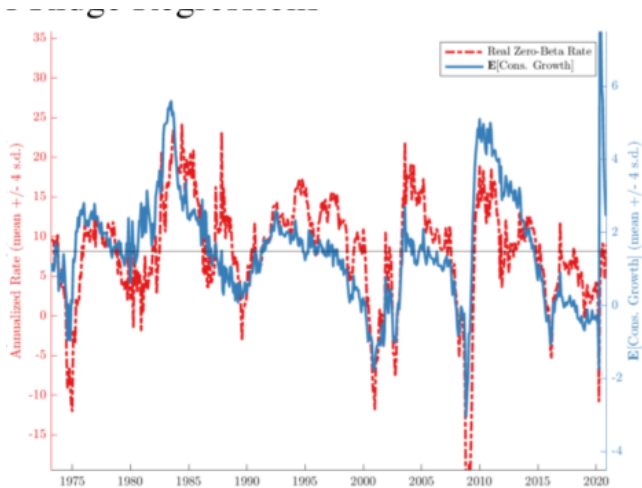
Treasury bill yield does not fit Euler equation [Back](#)



Better fit when omitting COVID

[Back](#)

Fit remains good even with ridge penalty [Back](#)



Predictive regressions [Back](#)

- $\mathbb{E}_t[\log(c_{t+1}/c_t)]$ and $\mathbb{E}_t[\log(P_{t+1}/P_t)]$ from predicted reg on instruments:

$$\log(c_{t+1}/c_t) = \gamma'_c Z_t + u_{c,t+1}$$

$$\log(P_{t+1}/P_t) = \gamma'_p Z_t + u_{p,t+1}$$

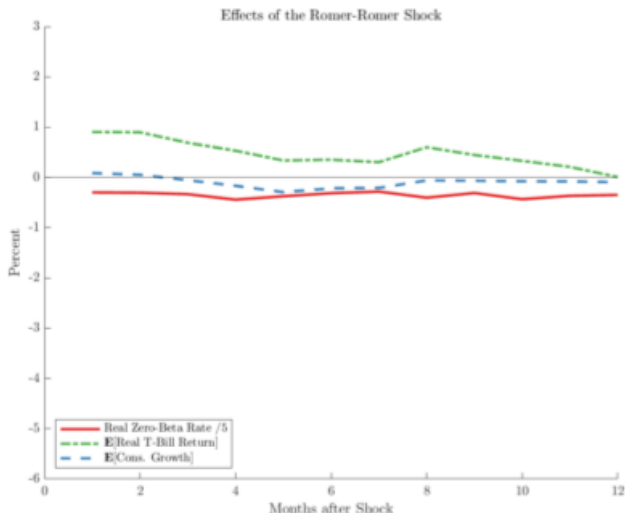
- Define: $\Delta = \gamma - \gamma_p - \sigma \times \gamma_c$
- Then graph shows: $\text{Var}(\Delta' Z_t) \approx 0$

- Let $\hat{V}_{Test}(\sigma)$ be the variance-covariance matrix of $\frac{1}{T} \sum_{t=1}^T g_{Test,t}(\hat{\theta}, \hat{\delta}(\sigma), \sigma)$, then under the null

$$\hat{S}(\sigma) = \left(\frac{1}{T} \sum_{t=1}^T g_{Test,t}(\hat{\theta}, \hat{\delta}(\sigma), \sigma) \right)' \hat{V}_{Test}(\sigma)^{-1} \left(\frac{1}{T} \sum_{t=1}^T g_{Test,t}(\hat{\theta}, \hat{\delta}(\sigma), \sigma) \right)$$

is chi-square with L degrees of freedom

Monetary tightening lowers zero beta rate (RR shock) [Back](#)



Monetary tightening lowers zero beta rate [Back](#)

- Run 6-month changes regression using coefficients from predictive regression:

$$\hat{\gamma}_l(Z_{l,t+5} - Z_{l,t-1}) = \phi_0^l + \phi_1^l mpshock_t + \varepsilon_{t+5}^l$$

Table 4: Decomposition of the Effects of a Nakamura-Steinsson Shock

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Real Bill	Real Z.B.	Ex. C. Gr.	g_RF	g_UMP	g_EBP	g_TSP	g_CPI_Rolling
NS_shock	1.801	-9.469	-1.873	6.107	-2.272	-3.256	-6.765	-0.886
Constant	-0.0856	-0.00218	0.00492	-0.132	-0.00881	0.167	-0.0375	0.00373
Observations	230	230	230	230	230	230	230	230

More stuff [Back](#)

- Risk premium is low (zero?) [Implied PD ratio](#)
- Aggregate EE does not need individual households to be on their EE (Werning 2015)
 - ▶ Of course it is possible to write models where aggregate EE fails, but this paper is evidence against such models
- What can go wrong in estimation? [More estimation](#)
- What if there are multiple SDFs? [More SDF](#)
- Other evidence that $R_{0,t}$ is correct intertemporal price [Banks, FX, TANK](#)

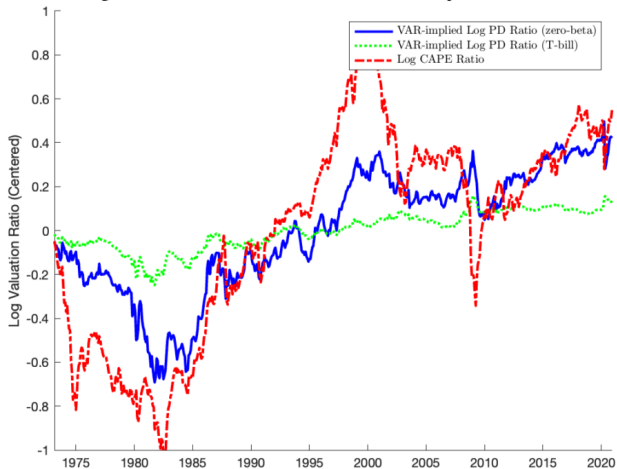
Is there an equity risk premium? [Back](#)

- Point estimate imply no to negative risk premium after controlling for consumption risk
 - ▶ Solve equity premium puzzle
 - ▶ Introduce "convenience puzzle"
- Cochrane 2011: variation in expected equity returns must explain volatility of valuation ratios

$$\mathbb{E}_t[pd_t] = (\sigma^{-1} - 1)\gamma'_0(I - \rho\Phi)^{-1}Z_t + cons$$

Is there an equity risk premium? [Back](#)

Figure 6: Predicted Valuation Ratio of a Consumption Claim



What can go wrong in estimation? [Back](#)

- Too many factors? unbiased but noisy
- Too few factors? zero beta portfolio may contain omitted risk premium: either biased δ , σ or EE fails
- Too few predictors? EE fails
- Too many predictors? Possibly weak instruments

What if there are multiple SDFs? [Back](#)

- Assumption is Λ_t is spanned by factors F_t
- Even so, for another $\tilde{R}_{0,t}$, there is an SDF $\tilde{\Lambda}_t$ that prices stocks $\mathbb{E}_t[\tilde{\Lambda}_{t+1}/\tilde{\Lambda}_t \times R_{i,t+1}] = 1$ and $\tilde{R}_{0,t} = \mathbb{E}_t[\tilde{\Lambda}_{t+1}/\tilde{\Lambda}_t]^{-1}$
- Then $R_{0,t} - \tilde{R}_{0,t}$ due to omitted factor
- But then vector of 1s spans betas to additional factor
- Procedure in text: recover $R_{0,t}$ that does not require additional factor to which all assets are exposed

Other evidence that $R_{0,t}$ is correct intertemporal price [Back](#)

- Banks favor debt over equity financing
 - ▶ Because zero beta rates are higher than Treasury bills
 - ▶ Arbitrage opportunities like CIP deviations (Du et al 2018)
- Disconnect between exchange rates and macroeconomy
 - ▶ Instead of adding wedges, there is a spread between bond yields and zero beta rates
- TANK models distinguish liquid and illiquid assets